

Tianyu (Terry) Zhang

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Education

The University of Texas at Austin Masters in Science in Data Science August 2026 - May 2028
Austin, Texas

The University of Texas at Austin Bachelor of Science in Mechanical Engineering August 2022 - May 2025
Austin, Texas Minor: Computer Science
Overall GPA: 3.75

Relevant Coursework: Data Structures & Algorithms, Databases, Data Analytics, Eng. Statistics and Finance

Skills: Python, SQL, Pandas, PyTorch, R, Docker, Java, JavaScript, HTML, CSS, Swift, MATLAB, Solidworks

Projects

Specialized RAG Pipeline - Python, PyTorch, FAISS, FastAPI, Docker June 2026 - August 2026

- Built a local retrieval augmented generation pipeline, retrieving external documents to answer user prompts
- Implemented an initial quick vector search on embedded chunks from 16 different SEC financial filings
- Trained a re-ranker in PyTorch to filter through initial search results, increasing top 5 search accuracy by >25%
- Labeled 7,300 training pairs from financial filings for re-ranker training, reaching 89% validation accuracy

Student Academics Prediction Modeling - pandas, scikit-learn June 2024 - August 2024

- Tested 6 machine learning models on 2,300+ student records to predict academic success with 94.6% accuracy
- Identified “absences” as the most negative correlation to grades and used this to predict GPA within 0.2 points
- Built an interactive web dashboard using HTML and SVG and displayed the final findings in 5 visual charts

Automated Plant Caretaker - Swift, Arduino, Solidworks August 2024 - December 2024

- Developed an iOS app for the remote control of sensors and motors satisfying 87% of surveyed users' requests
- Documented customer needs, DFM, DFA, along with other design principles in a detailed 35-page report

Experience

Thoroughfare LLC - Part-Time Data & Web Consultant April 2026 - August 2026

- Built a centralized Postgres database consisting of POS and sales data across 3 locations with nightly syncing
- Deployed the business's public website using Vercel increasing overall ordering volume by roughly 25%

Ford Audio and Video - Project Engineer; Austin, TX October 2025 - March 2026

- Produced AutoCAD floor plans, signal diagrams, and rack elevation diagrams across 2+ commercial AV projects
- Delivered 2 projects from start to finish by cooperating with PMs and installers resulting in \$20,000+ in revenue
- Performed field engineering and resolved 5+ termination errors, maintaining client system downtimes to <1 hour

Leadership and Activities

Student Engineers Educating Kids (SEEK) - VP Internal Relations, Program Officer August 2022 - May 2025

- Demoed 10 STEM projects for 6 semesters through afterschool visits to increase youth participation in STEM
- Created 3 original engineering projects for K-6 students by using analogies based on real engineering concepts

Additional Information

Languages: Fluent in English and Conversational Mandarin Chinese

Certifications: Google AI Essentials, FAA Private Pilots License

Interests: Aviation, Photography & Videography, Travel, Classical Music Performance

Work Eligibility: Eligible to work in the U.S. with no restrictions